SOME RESULTS ON DIMENSION SUBRINGS IN LIE RINGS

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1. Introduction

Let G be a group. Its lower central series is defined by

$$\gamma_1(G) = G, \qquad \gamma_{n+1}(G) = [\gamma_n(G), G].$$

Let $\Delta(G)$ denote the augmentation ideal of the integral group ring $\mathbb{Z}[G]$. The dimension subgroups of G are

$$D_n(G) = G \cap \big(1 + \Delta(G)^n\big) = \{g \in G \mid g - 1 \in \Delta(G)^n\}.$$

The commutator identity in $\mathbb{Z}[G]$ gives $\gamma_n(G) \subseteq D_n(G)$ for every n. The equality $D_n(G) = \gamma_n(G)$ holds for $n \leq 3$, but fails in general: Rips constructed a group for which $D_4(G) \neq \gamma_4(G)$ [**?**]. The free-group and free-Lie-ring cases are due to Magnus and Witt, respectively [**?**, **?**]. We refer to [**?**, **?**] for the classical dimension-subgroup problem and its subsequent development.

There is a parallel theory for Lie rings. For a Lie ring L, let $\mathrm{U}(L)$ be its universal enveloping algebra and let $\varpi(L)$ be the kernel of the augmentation $\mathrm{U}(L) \to \mathbb{Z}$. The lower central series and the dimension series of L are

$$\gamma_1(L) = L, \qquad \gamma_{n+1}(L) = [\gamma_n(L), L], \qquad \delta_n(L) = L \cap \varpi(L)^n.$$

Here and below L is identified with its image in $\mathrm{U}(L)$; this is valid over $\mathbb{Z}$ by the integral Birkhoff–Witt theorem. As in the group case,

$$\gamma_n(L) \subseteq \delta_n(L).$$

Bartholdi and Passi proved that $\delta_n(L) = \gamma_n(L)$ for $n \leq 3$ and constructed a Lie-ring analogue of the Rips example in degree four [**?**].

1.1. No stabilization for dimension series.

The lower central series has an elementary stabilization property: if $\gamma_m(L) = \gamma_{m+1}(L)$, then $\gamma_q(L) = \gamma_m(L)$ for every $q \geq m$. The analogous statement fails for the dimension series: arbitrarily many consecutive terms may coincide without subsequent stabilization.

Theorem A. *For every integer $N \geq 1$, there exist an integer $m = m(N) \geq 1$ and a Lie ring L_N such that*

$$\delta_m(L_N) = \delta_{m+1}(L_N) = \cdots = \delta_{m+N}(L_N) \neq \delta_{m+N+1}(L_N).$$

An explicit construction of L_N is given in Section 3. In this construction, $m(N) = N + 4$, and L_N is metabelian and nilpotent of class $N + 3$.

1.2. Centrality of dimension series.

Gupta and Kuz'min proved that, for every group G and every $n \geq 1$, $D_n(G)/\gamma_{n+1}(G)$ is abelian [**?**]; equivalently,

$$[D_n(G), D_n(G)] \subseteq \gamma_{n+1}(G).$$

Bartholdi and Passi proved that $\delta_n(L)/\gamma_{n+1}(L)$ is abelian for every Lie ring L [**?**]. Gupta and Kuz'min also proved that

$$[D_4(G), G] = \gamma_5(G)$$

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for every group G [?]. For general n, however, $D_n(G)/\gamma_{n+1}(G)$ need not be central in $G/\gamma_{n+1}(G)$ [?]. Passi and Sicking proved that

$$[\delta_n(L), L] = \gamma_{n+1}(L)$$

for every metabelian Lie ring L [?]. For a Lie algebra over a commutative ring k, define $\delta_n(L)$ as the inverse image of $\varpi_k(L)^n$ under the canonical map $L \to \mathrm{U}_k(L)$. The following theorem shows that the metabelian hypothesis is unnecessary.

Theorem B. Lean 4

Let L be a Lie algebra over a commutative ring with identity. For all $m, n \geq 1$,

$$[\delta_n(L), \delta_m(L)] \subseteq \gamma_{n+m}(L).$$

In particular,

$$[\delta_n(L), L] = \gamma_{n+1}(L).$$

1.3. **Plotkin's problem for metabelian Lie rings.** We prove a uniform vanishing bound for metabelian nilpotent Lie rings.

Theorem C. Lean 4

Let $c \geq 2$. If L is a metabelian Lie ring of nilpotency class at most c, then

$$\delta_{2c-1}(L) = 0.$$

Theorem C shows that the construction in Theorem A is optimal with respect to the nilpotency class: every metabelian Lie ring L of nilpotency class at most $N+3$ satisfies

$$\delta_{m(N)+N+1}(L) = 0, \qquad m(N) = N + 4.$$

Corollary D. Lean 4

For every Lie ring L and every $c \geq 2$,

$$\delta_{2c-1}(L) \subseteq \gamma_{c+1}(L) + L''.$$

In particular, $\delta_5(L) \subseteq \gamma_4(L)$.

The corresponding question for groups remains open: is $D_5(G) \subseteq \gamma_4(G)$ for every group G? See [?, Problem 20.84].

For a group G, put

$$D_\omega(G) = \bigcap_{n \geq 1} D_n(G), \qquad \gamma_\omega(G) = \bigcap_{n \geq 1} \gamma_n(G).$$

Plotkin asked whether $D_\omega(G) = \gamma_\omega(G)$ for every group G [?]. Our result gives an affirmative answer to the Lie-ring analogue of Plotkin's problem for metabelian Lie rings. For a Lie ring L, define analogously

$$\delta_\omega(L) = \bigcap_{n \geq 1} \delta_n(L), \qquad \gamma_\omega(L) = \bigcap_{n \geq 1} \gamma_n(L).$$

Corollary E. Lean 4

If L is a metabelian Lie ring, then

$$\delta_{2c-1}(L) \subseteq \gamma_{c+1}(L), \qquad c \geq 2.$$

In particular, $\delta_\omega(L) = \gamma_\omega(L)$.

1.4. **Organization.** Section 2 records the integral PBW theorem and the filtrations used below. Sections 3, 4, and 5 prove Theorems A, B, and C, respectively. Section 6 describes their verification in Lean 4.

2. Preliminaries

All commutators are left-normed:

$$[x_1, x_2, \ldots, x_r] = [[\cdots[[x_1, x_2], x_3], \ldots], x_r].$$

For additive subgroups $B_1, \ldots, B_r$ of a Lie ring, the notation $[B_1, \ldots, B_r]$ denotes the subgroup generated by the corresponding commutators. For an ideal A of L, set

$$[A, {}_0L] = A, \qquad [A, {}_{r+1}L] = [[A, {}_rL], L].$$

A Lie ring L is metabelian if $L'' = [[L, L], [L, L]] = 0$.

2.1. **The integral PBW theorem.** Let $T(L) = \bigoplus_{r\geq 0} L^{\otimes r}$ be the tensor algebra of the underlying abelian group of L. The universal enveloping algebra is

$$\mathrm{U}(L) = T(L)/(x \otimes y - y \otimes x - [x, y] \mid x, y \in L).$$

The tensor-length filtration induces the increasing PBW filtration

$$P_0\,\mathrm{U}(L) \subseteq P_1\,\mathrm{U}(L) \subseteq P_2\,\mathrm{U}(L) \subseteq \cdots,$$

where $P_r\,\mathrm{U}(L)$ is generated additively by products of at most r elements of L.

Theorem 2.1. *For every Lie ring L, the canonical maps*

$$L \longrightarrow \mathrm{U}(L), \qquad \mathrm{Sym}_{\mathbb{Z}}(L) \longrightarrow \mathrm{gr}_P\,\mathrm{U}(L)$$

are, respectively, injective and an isomorphism of graded algebras.

This is the Birkhoff–Witt theorem over the Dedekind domain $\mathbb{Z}$; see Higgins [**?**]. We use the following explicit consequence.

Corollary 2.2. *Suppose that the additive group of L has an ordered decomposition*

$$L = \bigoplus_{i\in I} \langle x_i \rangle,$$

where each summand is cyclic, finite or infinite. Then the following statements hold.

(i) For every $r \geq 1$, the quotient $P_r\,\mathrm{U}(L)/P_{r-1}\,\mathrm{U}(L)$ is the direct sum of the cyclic groups generated by the ordered products

$$x_{i_1} \cdots x_{i_r}, \qquad i_1 \leq \cdots \leq i_r.$$

The additive order of such a generator is the greatest common divisor of the additive orders of its factors, with order zero interpreted as infinite.

(ii) For $r \geq 1$, let M_r be the subgroup of $\mathrm{U}(L)$ generated by the ordered products of length r, and put $M_0 = \mathbb{Z} \cdot 1$. Then

$$P_r\,\mathrm{U}(L) = \bigoplus_{q=0}^{r} M_q, \qquad \mathrm{U}(L) = \bigoplus_{q\geq 0} M_q.$$

Consequently, for every $s \geq 0$, the rule

$$p_{\leq s}(x_{i_1} \cdots x_{i_r}) = \begin{cases} x_{i_1} \cdots x_{i_r}, & r \leq s, \\ 0, & r > s, \end{cases} \qquad i_1 \leq \cdots \leq i_r,$$

defines an additive retraction

$$p_{\leq s} : \mathrm{U}(L) \longrightarrow P_s\,\mathrm{U}(L).$$

Proof. By Theorem 2.1, the quotient in (i) is $\mathrm{Sym}^r_{\mathbb{Z}}(L)$. Thus (i) follows by taking symmetric powers of the given direct sum.

For (ii), part (i) shows that the map

$$M_r \longrightarrow P_r\,\mathrm{U}(L)/P_{r-1}\,\mathrm{U}(L)$$

is surjective. If an element of M_r maps to zero, the coefficient of each ordered product is divisible by the additive order of its image in the quotient. If this order is zero, the coefficient is zero. Otherwise, let d_j be the additive order of x_{i_j}, with $d_j = 0$ for infinite order, and set $d = \gcd(d_1, \dots, d_r)$. By (i), d is the order of the image of $x_{i_1} \cdots x_{i_r}$. Each d_j annihilates this product, so Bézout's identity implies that d also annihilates it. Therefore the map is injective, and $P_r \mathrm{U}(L) = P_{r-1} \mathrm{U}(L) \oplus M_r$. Induction on r proves both decompositions and the assertion about $p_{\le s}$. $\square$

The augmentation $\mathrm{U}(L) \to \mathbb{Z}$ is induced by the zero map $L \to \mathbb{Z}$. Its kernel $\varpi(L)$ is the two-sided ideal generated by L. Thus $\varpi(L)^n$ is generated additively by products of at least n elements of L.

2.2. **Dimension subrings.** For every $n \ge 1$,

$$\gamma_n(L) \subseteq \delta_n(L). \tag{1}$$

Indeed, $L \subseteq \varpi(L)$ and $[\varpi(L)^r, \varpi(L)] \subseteq \varpi(L)^{r+1}$, so (1) follows by induction. Both series are functorial: a homomorphism $f : L \to M$ satisfies

$$f(\gamma_n(L)) \subseteq \gamma_n(M), \qquad f(\delta_n(L)) \subseteq \delta_n(M).$$

We also use the following result of Passi and Sicking.

Theorem 2.3. *If L is metabelian, then, for every $n \ge 1$,*

$$2\delta_n(L) \subseteq \gamma_n(L).$$

This is [**?**, Theorem 1.1].

3. Finite plateaux

Fix an integer $N \ge 1$. Let F be the free metabelian Lie ring on $x_1, \dots, x_5$. Order the generators by

$$x_5 < x_1 < x_2 < x_3 < x_4.$$

The standard metabelian Hall basis consists of the generators and the commutators

$$[x_{i_0}, x_{i_1}, \dots, x_{i_r}], \qquad x_{i_0} > x_{i_1} \le x_{i_2} \le \cdots \le x_{i_r}.$$

Put

$$t = [x_4, \underbrace{x_5, \dots, x_5}_{N}], \qquad c_i = [t, x_i], \qquad u_i = [c_i, x_i], \quad 1 \le i \le 3.$$

Thus t, c_i, and u_i have weights $N+1$, $N+2$, and $N+3$, respectively. With the chosen order, the three u_i are distinct Hall basis elements.

Define the relators

$$\begin{aligned} r_1 &= 4x_1 + 2c_3 + c_2, \\ r_2 &= 16x_2 + 4c_3 - c_1, \\ r_3 &= 64x_3 - 4c_2 - 2c_1. \end{aligned} \tag{2}$$

Let L_N be the quotient of F by the Lie ideal specified as follows:

(i) impose $r_1 = r_2 = r_3 = 0$;
(ii) factor out $\gamma_{N+4}(F)$;
(iii) in weight $N+3$, factor out all Hall basis elements other than u_1, u_2, u_3;
(iv) impose

$$u_1 = 16u_3, \qquad u_2 = 4u_3, \qquad 64u_3 = 0.$$

Example 3.1. Let $\mathcal{H}_q$ denote the weight-q part of the metabelian Hall basis fixed above. A subscript met indicates a presentation in the variety of metabelian Lie rings. Substituting $N = 1$ gives

$$L_1 = \left\langle x_1, x_2, x_3, x_4, x_5 \;\middle|\; \begin{array}{ll} \text{(i)} & \begin{array}{c} 4x_1 + 2[x_4, x_5, x_3] + [x_4, x_5, x_2] = 0, \\ 16x_2 + 4[x_4, x_5, x_3] - [x_4, x_5, x_1] = 0, \\ 64x_3 - 4[x_4, x_5, x_2] - 2[x_4, x_5, x_1] = 0; \end{array} \\ \text{(ii)} & \gamma_5 = 0; \\ \text{(iii)} & \begin{array}{r} h = 0 \quad \text{for } h \in \mathcal{H}_4 \setminus \{[x_4, x_5, x_1, x_1], \\ [x_4, x_5, x_2, x_2], [x_4, x_5, x_3, x_3]\}; \end{array} \\ \text{(iv)} & \begin{array}{c} [x_4, x_5, x_1, x_1] = 16[x_4, x_5, x_3, x_3], \\ [x_4, x_5, x_2, x_2] = 4[x_4, x_5, x_3, x_3], \\ 64[x_4, x_5, x_3, x_3] = 0. \end{array} \end{array} \right\rangle_{\text{met}}.$$

Similarly,

$$L_2 = \left\langle x_1, x_2, x_3, x_4, x_5 \;\middle|\; \begin{array}{ll} \text{(i)} & \begin{array}{c} 4x_1 + 2[x_4, x_5, x_5, x_3] + [x_4, x_5, x_5, x_2] = 0, \\ 16x_2 + 4[x_4, x_5, x_5, x_3] - [x_4, x_5, x_5, x_1] = 0, \\ 64x_3 - 4[x_4, x_5, x_5, x_2] - 2[x_4, x_5, x_5, x_1] = 0; \end{array} \\ \text{(ii)} & \gamma_6 = 0; \\ \text{(iii)} & \begin{array}{r} h = 0 \quad \text{for } h \in \mathcal{H}_5 \setminus \{[x_4, x_5, x_5, x_1, x_1], \\ [x_4, x_5, x_5, x_2, x_2], [x_4, x_5, x_5, x_3, x_3]\}; \end{array} \\ \text{(iv)} & \begin{array}{c} [x_4, x_5, x_5, x_1, x_1] = 16[x_4, x_5, x_5, x_3, x_3], \\ [x_4, x_5, x_5, x_2, x_2] = 4[x_4, x_5, x_5, x_3, x_3], \\ 64[x_4, x_5, x_5, x_3, x_3] = 0. \end{array} \end{array} \right\rangle_{\text{met}}.$$

Lemma 3.2. *The Lie ring L_N is metabelian and nilpotent of class $N + 3$, and*

$$\gamma_{N+3}(L_N) = \langle u_3 \rangle \cong \mathbb{Z}/64.$$

Proof. Metabelianity is inherited from F, and relation (ii) gives $\gamma_{N+4}(L_N) = 0$. By relation (iii), $\gamma_{N+3}(L_N)$ is generated by u_1, u_2, u_3. Hall collection shows that the induced relations are precisely those in (iv), with matrix

$$\begin{pmatrix} 1 & 0 & -16 \\ 0 & 1 & -4 \\ 0 & 0 & 64 \end{pmatrix}.$$

Its Smith reduction gives a cyclic quotient of order 64, generated by u_3. Thus $\gamma_{N+3}(L_N) \neq 0$, while $\gamma_{N+4}(L_N) = 0$. □

Set

$$b_{12} = [x_1, x_2], \qquad b_{13} = [x_1, x_3], \qquad b_{23} = [x_2, x_3]$$

and define

$$a = 32b_{12} + 64b_{13} + 128b_{23} \in L_N. \tag{3}$$

Bracketing the relators (2) with x_1, x_2, x_3 and using $[F', F'] = 0$ together with relation (iii) gives

$$\begin{array}{rclrcl} 4b_{12} + u_2 & = & 0, & 4b_{13} + 2u_3 & = & 0, \\ -16b_{12} - u_1 & = & 0, & 16b_{23} + 4u_3 & = & 0, \\ -64b_{13} - 2u_1 & = & 0, & -64b_{23} - 4u_2 & = & 0. \end{array} \tag{4}$$

Lemma 3.3. *In L_N one has $a = 32u_3$. In particular,*

$$\langle a \rangle \cong \mathbb{Z}/2.$$

Proof. Three relations in (4) give

$$32b_{12} = -2u_1, \qquad 64b_{13} = -2u_1, \qquad 128b_{23} = -8u_2.$$

Since $u_1 = 16u_3$, $u_2 = 4u_3$, and $64u_3 = 0$, it follows that

$$a = -4u_1 - 8u_2 = -96u_3 = 32u_3.$$

By Lemma 3.2, the element u_3 has order 64; hence $32u_3$ has order 2. $\square$

Lemma 3.4. *The element a belongs to $\delta_{2N+4}(L_N)$.*

Proof. In $\mathrm{U}(L_N)$, expand (3) as

$$a = x_1w_1 + x_2w_2 + x_3w_3,$$

where

$$w_1 = 32x_2 + 64x_3, \qquad w_2 = -32x_1 + 128x_3, \qquad w_3 = -64x_1 - 128x_2.$$

Since $t, c_i \in F'$ and F is metabelian, $[t, c_i] = 0$. Bracketing the relators (2) with t gives

$$4c_1 = 0, \qquad 16c_2 = 0, \qquad 64c_3 = 0.$$

Together with the relators themselves, these identities give

$$w_1 = 4(c_1 + c_2 - 2c_3), \qquad w_2 = 16(c_2 + c_3), \qquad w_3 = 64c_3.$$

Substitution from (2) gives

$$4x_1 = -(2c_3 + c_2), \qquad 16x_2 = -(4c_3 - c_1), \qquad 64x_3 = 4c_2 + 2c_1.$$

The elements c_1, c_2, c_3 lie in the abelian derived ring and therefore commute in $\mathrm{U}(L_N)$. Substituting the preceding identities gives

$$\begin{aligned} a &= (4x_1)(c_1 + c_2 - 2c_3) + (16x_2)(c_2 + c_3) + (64x_3)c_3 \\ &= c_1c_3 - c_2^2. \end{aligned} \tag{5}$$

Since $c_i \in \gamma_{N+2}(L_N) \subseteq \varpi(L_N)^{N+2}$, equation (5) implies $a \in \varpi(L_N)^{2N+4}$. $\square$

Lemma 3.5.

$$L_N \cap \varpi(L_N)^{N+4} = \langle a \rangle.$$

Proof. Let $z \in L_N \cap \varpi(L_N)^{N+4}$. Since $\gamma_{N+4}(L_N) = 0$, Theorem 2.3 gives $2z = 0$. Theorem B also gives $[z, L_N] = \gamma_{N+5}(L_N) = 0$. Express z in the fixed metabelian Hall basis and apply PBW collection to its image in $\mathrm{U}(L_N)$. In degrees below $N+4$, the Hall elements of weights $3, \dots, N+2$ have distinct leading PBW monomials, so their coefficients vanish. The weight-two components of the equations $[z, x_i] = 0$, together with (2), leave only a linear combination of

$$b_{12}, \quad b_{13}, \quad b_{23}$$

together with u_3.

After substituting $u_1 = 16u_3$ and $u_2 = 4u_3$ into (4) and discarding redundant relations, a relation matrix for these four terms, with columns ordered as $(b_{12}, b_{13}, b_{23}, u_3)$, is

$$\begin{pmatrix} 4 & 0 & 0 & 4 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 16 & 4 \\ 0 & 0 & 0 & 64 \end{pmatrix}. \tag{6}$$

Let

$$v = \alpha b_{12} + \beta b_{13} + \chi b_{23} + \eta u_3.$$

Corollary 2.2, applied in degrees 2 and $N+3$, shows that v has trivial image in $\mathrm{U}(L_N)/\varpi(L_N)^{N+4}$ precisely when

$$\alpha \equiv 0 \pmod 4, \qquad \beta \equiv 0 \pmod 4, \qquad \chi \equiv 0 \pmod{16},$$

and

$$\eta - \alpha - \frac{\beta}{2} - \frac{\chi}{4} \equiv 0 \pmod{32}.$$

Under the first three congruences, (6) gives

$$v = \left(\eta - \alpha - \frac{\beta}{2} - \frac{\chi}{4}\right) u_3.$$

Thus the kernel is generated by $32u_3 = a$, and it has order two by Lemma 3.3. □

Lemma 3.6. *The element a does not belong to $\varpi(L_N)^{2N+5}$.*

Proof. Consider the augmentation-graded algebra

$$\mathrm{gr}_{\varpi}\, \mathrm{U}(L_N) = \bigoplus_{q \geq 0} \varpi(L_N)^q / \varpi(L_N)^{q+1}.$$

By (5), the initial form of a in degree $2N+4$ is

$$\bar{c}_1 \bar{c}_3 - \bar{c}_2^2.$$

The initial forms of the defining relators give

$$4\bar{c}_1 = 0, \qquad 16\bar{c}_2 = 0, \qquad 64\bar{c}_3 = 0.$$

Choose the PBW order so that $\bar{c}_1\bar{c}_3$ is an ordered monomial. PBW collection in degree $2N+4$ shows that every relation in this graded component has even coefficient at $\bar{c}_1\bar{c}_3$. Therefore the rule

$$\varepsilon(\bar{c}_1\bar{c}_3) = 1, \qquad \varepsilon(M) = 0 \quad \text{for every other ordered PBW monomial } M \text{ of this degree}$$

is well defined and gives a homomorphism

$$\varepsilon : \varpi(L_N)^{2N+4} / \varpi(L_N)^{2N+5} \longrightarrow \mathbb{Z}/2.$$

It sends the initial form of a to 1. That initial form is therefore nonzero. □

Proof of Theorem A. Let $N \geq 1$ and take the Lie ring L_N constructed above. If $q \geq N+4$, then Lemma 3.5 gives

$$\delta_q(L_N) \subseteq \delta_{N+4}(L_N) = \langle a \rangle.$$

For $N+4 \leq q \leq 2N+4$, Lemma 3.4 gives

$$\langle a \rangle \subseteq \delta_{2N+4}(L_N) \subseteq \delta_q(L_N).$$

Consequently,

$$\delta_{N+4}(L_N) = \cdots = \delta_{2N+4}(L_N) = \langle a \rangle \cong \mathbb{Z}/2.$$

Here the last isomorphism follows from Lemma 3.3. Finally, $\delta_{2N+5}(L_N) \subseteq \langle a \rangle$, whereas Lemma 3.6 gives $a \notin \delta_{2N+5}(L_N)$. Hence $\delta_{2N+5}(L_N) = 0$. The nilpotency statement follows from Lemma 3.2. □

4. Dimension centrality

Throughout this section, k is a commutative ring with identity and L is a Lie algebra over k. Write $\iota : L \to \mathrm{U}_k(L)$ for the canonical map and $\varpi_k(L)$ for the augmentation ideal. We set

$$\delta_n(L) = \iota^{-1}(\varpi_k(L)^n).$$

The map ι is not assumed to be injective.

Lemma 4.1. *Let A be an ideal of L. The adjoint action extends uniquely to a right $\mathrm{U}_k(L)$-module structure on A, and*

$$A \cdot \varpi_k(L)^n = [A, {}_n L].$$

Proof. On the tensor algebra of L, define

$$a \cdot 1 = a, \qquad a \cdot (x_1 \cdots x_r) = [a, x_1, \ldots, x_r].$$

For $a \in A$ and $x, y \in L$, the Jacobi identity gives

$$[[a, x], y] - [[a, y], x] - [a, [x, y]] = 0.$$

Hence the defining relations of $\mathrm{U}_k(L)$ act trivially, and the action factors through $\mathrm{U}_k(L)$. The ideal $\varpi_k(L)^n$ is spanned over k by products of at least n elements of L. A product of length $r \geq n$

acts into $[A, {}_r L] \subseteq [A, {}_n L]$. Conversely, every generator $[a, x_1, \ldots, x_n]$ of $[A, {}_n L]$ is the action of $x_1 \cdots x_n \in \varpi_k(L)^n$ on a. $\square$

Proof of Theorem B. Let A be an ideal of L, let $a \in A$, and let $x \in \delta_r(L)$. Lemma 4.1 gives

$$[a, x] = a \cdot \iota(x) \in A \cdot \varpi_k(L)^r = [A, {}_r L].$$

Thus $[A, \delta_r(L)] \subseteq [A, {}_r L]$. Taking $A = L$ and using skew-symmetry gives $[\delta_r(L), L] \subseteq \gamma_{r+1}(L)$. The inclusion $\gamma_r(L) \subseteq \delta_r(L)$ gives the reverse containment:

$$\gamma_{r+1}(L) = [\gamma_r(L), L] \subseteq [\delta_r(L), L].$$

Hence $[\delta_r(L), L] = \gamma_{r+1}(L)$. Since $\delta_n(L)$ is the inverse image of the two-sided ideal $\varpi_k(L)^n$, it is an ideal of L. Applying the relative inclusion with $A = \delta_n(L)$ and $r = m$ gives

$$\begin{aligned}[\delta_n(L), \delta_m(L)] &\subseteq [\delta_n(L), {}_m L] \\ &= [\gamma_{n+1}(L), {}_{m-1} L] = \gamma_{n+m}(L).\end{aligned}$$

$\square$

5. Metabelian nilpotent Lie rings

We return to integral Lie rings. We first reduce Theorem C to the finitely generated case.

Proposition 5.1. *Let $c \geq 2$. If L is a Lie ring such that*

$$\delta_{2c-1}(L) \neq 0,$$

then there is a finitely generated Lie subring $L_0 \subseteq L$ such that

$$\delta_{2c-1}(L_0) \neq 0.$$

Consequently, the general case of Theorem C follows from the finitely generated case.

Proof. Write L as the filtered union of its finitely generated Lie subrings. The universal enveloping algebra functor commutes with filtered colimits, as does each fixed power of the augmentation ideal. Hence

$$\mathrm{U}(L) = \varinjlim_{A \subseteq L} \mathrm{U}(A), \qquad \varpi(L)^{2c-1} = \varinjlim_{A \subseteq L} \varpi(A)^{2c-1},$$

where A runs over the finitely generated Lie subrings of L. If $0 \neq z \in L \cap \varpi(L)^{2c-1}$, the two displayed colimits give a finitely generated Lie subring $L_0 \subseteq L$ containing z such that $z \in \varpi(L_0)^{2c-1}$. Thus $0 \neq z \in \delta_{2c-1}(L_0)$. If L is metabelian of nilpotency class at most c, then so is L_0. $\square$

The proof of Theorem C proceeds as follows.

- By Proposition 5.1, it suffices to consider a finitely generated metabelian Lie ring L of nilpotency class at most c.
- We argue by contradiction. If $0 \neq z \in \delta_{2c-1}(L)$, there is a homomorphism $\langle z \rangle \to \mathbb{Q}/\mathbb{Z}$ that is nonzero on z. Since divisible abelian groups are injective, it extends to an additive homomorphism $\ell : L \to \mathbb{Q}/\mathbb{Z}$.
- We construct an additive extension $\Lambda : \mathrm{U}(L) \to \mathbb{Q}/\mathbb{Z}$ of ℓ such that $\Lambda(\varpi(L)^{2c-1}) = 0$. Since $z \in \varpi(L)^{2c-1}$, this gives $0 = \Lambda(z) = \ell(z)$, a contradiction.

Thus Theorem C follows from the following extension theorem.

Theorem 5.2. *Let $c \geq 2$, and let L be a finitely generated metabelian Lie ring of nilpotency class at most c. For every additive homomorphism $\ell : L \to \mathbb{Q}/\mathbb{Z}$, there is an additive homomorphism $\Lambda : \mathrm{U}(L) \to \mathbb{Q}/\mathbb{Z}$ such that*

$$\Lambda|_L = \ell, \qquad \Lambda(\varpi(L)^{2c-1}) = 0.$$

Step I: An auxiliary Lie ring. Put $L' = [L, L]$ and $L_{\mathrm{ab}} = L/L'$. Lie monomials of weights at most c in a finite Lie generating set generate the underlying abelian group of L. Thus L, L', and L_{ab} are finitely generated as abelian groups. Write L_{ab} in Smith normal form:

$$\begin{aligned} L_{\mathrm{ab}} &\cong \bigoplus_{i=1}^{k} \mathbb{Z}/d_i\mathbb{Z} \\ &= \langle e_1, \dots, e_k \mid d_i e_i = 0,\ 1 \le i \le k \rangle_{\mathrm{Ab}}, \qquad d_1 \mid d_2 \mid \cdots \mid d_k. \end{aligned}$$

Here $d_i \ge 0$, and $\mathbb{Z}/0\mathbb{Z}$ is understood to be $\mathbb{Z}$. Put

$$P = \bigoplus_{i=1}^{k} \mathbb{Z}e_i, \qquad K = \langle d_i e_i \mid 1 \le i \le k,\ d_i \ne 0 \rangle \subseteq P.$$

Then there is a short exact sequence of abelian groups

$$0 \longrightarrow K \longrightarrow P \xrightarrow{\rho} L_{\mathrm{ab}} \longrightarrow 0. \tag{7}$$

Endow P with the structure of a Lie ring with zero bracket and form the pullback

$$\widetilde{L} = L \times_{L_{\mathrm{ab}}} P = \{(x, p) \in L \times P \mid x + L' = \rho(p)\}.$$

Thus the bracket in $\widetilde{L}$ is

$$[(x, p), (y, q)] = ([x, y], 0).$$

Then $\pi : \widetilde{L} \to L$ is a central extension whose kernel is naturally identified with K:

$$\begin{tikzcd} 0 \arrow[r] & K \arrow[r] \arrow[d, equal] & \widetilde{L} \arrow[r, "\pi"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & L \arrow[r] \arrow[d, two heads] & 0 \\ 0 \arrow[r] & K \arrow[r] & P \arrow[r, "\rho"'] & L_{\mathrm{ab}} \arrow[r] & 0. \end{tikzcd}$$

The construction has the following properties:
(a) $K \subseteq Z(\widetilde{L})$;
(b) the map π induces isomorphisms

$$\pi : \gamma_j(\widetilde{L}) \xrightarrow{\cong} \gamma_j(L), \qquad j \ge 2.$$

In particular, $\widetilde{L}' = L'$, $\widetilde{L}'' = 0$, and $\gamma_{c+1}(\widetilde{L}) = 0$;
(c)

$$\ker(\pi_* : \mathrm{U}(\widetilde{L}) \longrightarrow \mathrm{U}(L)) = K\,\mathrm{U}(\widetilde{L}).$$

Properties (a) and (b) follow from the displayed bracket formula; property (c) follows from the universal property of the enveloping algebra. Consequently, to prove Theorem 5.2, it suffices, for each additive homomorphism $\ell : L \to \mathbb{Q}/\mathbb{Z}$, to construct an additive homomorphism $\widetilde{\Lambda} : \mathrm{U}(\widetilde{L}) \to \mathbb{Q}/\mathbb{Z}$ such that

$$\widetilde{\Lambda}|_{\widetilde{L}} = \ell \circ \pi, \qquad \widetilde{\Lambda}(K\,\mathrm{U}(\widetilde{L})) = 0, \qquad \widetilde{\Lambda}(\varpi(\widetilde{L})^{2c-1}) = 0. \tag{8}$$

Lemma 5.3. *The second projection $\widetilde{L} \to P$ admits an additive section. Consequently, there is an isomorphism of abelian groups*

$$\widetilde{L} \cong L' \oplus P.$$

Proof. Since P is free, choose an additive lift $s : P \to L$ of ρ, so that

$$s(p) + L' = \rho(p), \qquad p \in P.$$

The map

$$\sigma : P \longrightarrow \widetilde{L}, \qquad \sigma(p) = (s(p), p),$$

is a section of the second projection. The kernel of that projection is L', embedded by $a \mapsto (a, 0)$. Thus $\widetilde{L} = L' \oplus \sigma(P)$ as an abelian group. $\square$

We use the following consequence. For $x, y \in \widetilde{L}$ and $d \in \mathbb{Z}$,

$$dx, dy \in \widetilde{L}' + Z(\widetilde{L}) \implies d[x,y] \in Z(\widetilde{L}). \tag{9}$$

Indeed, for $z \in \widetilde{L}$, metabelianity and the Jacobi identity give

$$[d[x,y], z] = [dx, y, z] = [dx, z, y] = [x, z, dy] = 0.$$

Step II: A compatible retraction of the universal enveloping algebra.

Lemma 5.4. *There is an additive retraction*

$$p_{\leq 2} : \mathrm{U}(\widetilde{L}) \longrightarrow P_2\,\mathrm{U}(\widetilde{L})$$

such that

$$p_{\leq 2}(K\,\mathrm{U}(\widetilde{L})) \subseteq K + K\widetilde{L} + Z(\widetilde{L})\widetilde{L}.$$

Proof. Use the section σ from Lemma 5.3 to identify P with $\sigma(P)$, so that $\widetilde{L} = L' \oplus P$. Choose a cyclic decomposition of L'. Let $\mathcal{B}$ consist of the chosen generators of its cyclic summands, followed by $e_1 < \cdots < e_k$. Let $p_{\leq 2}$ be the additive retraction associated with this ordered decomposition by Corollary 2.2(ii).

Let $e = e_i$, let $d = d_i \neq 0$, and put $k = (0, d_i e_i) \in K$. Under the chosen decomposition of $\widetilde{L}$, one has $k = de + b$ for some $b \in L'$. It suffices to consider ku, where u is an ordered product of length r. For $a \in L'$,

$$d[e,a] = [de, a] = [de + b, a] = [k, a] = 0,$$

while for $f = e_j < e_i$ one has $d_j \mid d$ and

$$de = k - b \in \widetilde{L}' + K, \qquad d\rho(e_j) = 0 \implies df \in \widetilde{L}' + K.$$

Since $K \subseteq Z(\widetilde{L})$, (9) gives $d[e,f] \in Z(\widetilde{L})$. To put deu in PBW order, use

$$dea = dae \quad (a \in L'), \qquad def = dfe + d[e,f] \quad (f < e).$$

Thus deu becomes the sum of one ordered product of length $r+1$ and terms $d[e,f]v$, where v is an ordered product of length $r-1$. Since $d[e,f]$ is central, reordering these terms produces no additional terms. The product bu has length $r+1$, so

$$p_{\leq 2}(ku) \in \begin{cases} K + K\widetilde{L}, & r \leq 1, \\ Z(\widetilde{L})\widetilde{L}, & r = 2, \\ 0, & r > 2. \end{cases}$$

$\square$

Lemma 5.5. *If $r \geq 1$ and w is a product of r elements of $\widetilde{L}$, then*

$$p_{\leq 2}(w) \in \gamma_r(\widetilde{L}) + \sum_{0<i<r} \gamma_i(\widetilde{L})\gamma_{r-i}(\widetilde{L}).$$

Proof. Repeated application of $xy = yx + [x,y]$ expresses w as a sum of ordered products $u_1 \cdots u_q$, where each u_j is an iterated commutator and the sum of their weights is r. The map $p_{\leq 2}$ annihilates the terms with $q > 2$. A term with $q = 1$ belongs to $\gamma_r(\widetilde{L})$, and a term with $q = 2$ belongs to $\gamma_i(\widetilde{L})\gamma_{r-i}(\widetilde{L})$ for some $0 < i < r$. $\square$

Step III: Construction of an additive character.

Proposition 5.6. *Let $\varphi : \widetilde{L} \to \mathbb{Q}/\mathbb{Z}$ be an additive homomorphism satisfying $\varphi(K) = 0$. Then there is an additive homomorphism $\Phi : \mathrm{U}(\widetilde{L}) \to \mathbb{Q}/\mathbb{Z}$ such that*

(1) $\Phi|_{\widetilde{L}} = \varphi$;
(2) $\Phi(K\,\mathrm{U}(\widetilde{L})) = 0$;
(3) $\Phi(\varpi(\widetilde{L})^{2c-1}) = 0$.

Proof. By Lemma 5.3, the abelian group $\widetilde{L}$ is finitely generated; hence so is $\widetilde{L}/Z(\widetilde{L})$. Choose a cyclic decomposition

$$\widetilde{L}/Z(\widetilde{L}) = \bigoplus_{i=1}^{s} \langle \bar{v}_i \rangle$$

and lifts $v_i \in \widetilde{L}$. Write $\bar{x}$ for the image of $x \in \widetilde{L}$ in this quotient. Define

$$H(\bar{v}_i, \bar{v}_j) = \begin{cases} \varphi([v_i, v_j]), & i > j, \\ 0, & i \leq j, \end{cases}$$

and extend biadditively. If $n\bar{v}_i = 0$, then nv_i is central, so $n\varphi([v_i, v_j]) = \varphi([nv_i, v_j]) = 0$. The same argument in the second variable proves that H respects all cyclic relations. Thus H is well defined and satisfies

$$H(\bar{x}, \bar{y}) - H(\bar{y}, \bar{x}) = \varphi([x, y]). \tag{10}$$

Define

$$\lambda_2 : P_2 \, \mathrm{U}(\widetilde{L}) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

by

$$\lambda_2(1) = 0, \qquad \lambda_2(x) = \varphi(x), \qquad \lambda_2(xy) = H(\bar{x}, \bar{y}).$$

Biadditivity shows that the formulas respect the cyclic relations, and (10) gives

$$\lambda_2(xy - yx) = \lambda_2([x, y]).$$

Hence these formulas define λ_2. Put

$$\Phi = \lambda_2 p_{\leq 2} : \mathrm{U}(\widetilde{L}) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Since $p_{\leq 2}$ is a retraction, the restriction of Φ to $\widetilde{L}$ is φ. This proves (1).

For (2), the assumption $\varphi(K) = 0$ and the centrality of K give

$$\lambda_2(K) = \lambda_2(K\widetilde{L}) = \lambda_2(Z(\widetilde{L})\widetilde{L}) = 0.$$

Lemma 5.4 now yields $\Phi(K \, \mathrm{U}(\widetilde{L})) = 0$.

For (3), property (b) gives $\gamma_{c+1}(\widetilde{L}) = 0$, so $\gamma_c(\widetilde{L})$ is central. The augmentation power is spanned additively by products w of length $r \geq 2c - 1$. For each such w, Lemma 5.5 gives

$$p_{\leq 2}(w) \in \gamma_r(\widetilde{L}) + \sum_{0<i<r} \gamma_i(\widetilde{L})\gamma_{r-i}(\widetilde{L}).$$

Since $r \geq 2c - 1 \geq c + 1$, the first summand is zero. In each product in the sum, either $i \geq c$ or $r - i \geq c$. The corresponding factor lies in $\gamma_c(\widetilde{L})$ and is therefore central. Its image in $\widetilde{L}/Z(\widetilde{L})$ is zero, so λ_2 vanishes on the product. Hence Φ vanishes on every product of length at least $2c - 1$. ☐

Step IV: Proof of Theorem C and its consequences.

Proof of Theorem 5.2. Apply Proposition 5.6 to $\varphi = \ell \circ \pi$. Since $\pi(K) = 0$, its hypothesis is satisfied. Put $\widetilde{\Lambda} = \Phi$. Then the conditions in (8) hold. Property (c) and (8) show that $\widetilde{\Lambda}$ descends to an additive homomorphism $\Lambda : \mathrm{U}(L) \to \mathbb{Q}/\mathbb{Z}$ with $\Lambda|_L = \ell$. The quotient map sends $\varpi(\widetilde{L})^{2c-1}$ onto $\varpi(L)^{2c-1}$. Hence (8) gives $\Lambda(\varpi(L)^{2c-1}) = 0$. ☐

The construction is summarized in the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & K \arrow[r] & \widetilde{L} \arrow[r, "\pi"] \arrow[d] & L \arrow[r] \arrow[d] \arrow[ddrr, "\ell"] & 0 & \\
0 \arrow[r] & K\,\mathrm{U}(\widetilde{L}) \arrow[r, "\text{property (c)}"] \arrow[d, "p_{\leq 2}|_{K\,\mathrm{U}(\widetilde{L})}"', "\text{Lm. 5.4}"] & \mathrm{U}(\widetilde{L}) \arrow[r, "\pi_*"] \arrow[d, "p_{\leq 2}"] \arrow[drrr, "\widetilde{\Lambda}", "\text{Prop. 5.6}"'] & \mathrm{U}(L) \arrow[r] \arrow[drr, dashed, "\Lambda"] & 0 & \\
& K + K\widetilde{L} + Z(\widetilde{L})\widetilde{L} \arrow[r, hook] & P_2\,\mathrm{U}(\widetilde{L}) \arrow[rrr, "\lambda_2"'] & & & \mathbb{Q}/\mathbb{Z}.
\end{tikzcd}$$

Proof of Theorem C. By Proposition 5.1, it suffices to consider the finitely generated case. Suppose that $0 \neq z \in \delta_{2c-1}(L)$. A homomorphism $\langle z \rangle \to \mathbb{Q}/\mathbb{Z}$ nonzero on z extends to an additive homomorphism $\ell : L \to \mathbb{Q}/\mathbb{Z}$, since divisible abelian groups are injective. Theorem 5.2 gives $\Lambda : \mathrm{U}(L) \to \mathbb{Q}/\mathbb{Z}$ with $\Lambda|_L = \ell$ and $\Lambda(\varpi(L)^{2c-1}) = 0$. Thus $0 = \Lambda(z) = \ell(z)$, a contradiction. □

Corollary 5.7. Lean 4
For every Lie ring L and every $c \geq 2$,

$$\delta_{2c-1}(L) \subseteq \gamma_{c+1}(L) + L''.$$

In particular, $\delta_5(L) \subseteq \gamma_4(L)$.

Proof. Let $q : L \to L/(\gamma_{c+1}(L) + L'')$ be the quotient map. Its codomain is metabelian and has nilpotency class at most c, so Theorem C gives

$$\delta_{2c-1}(L/(\gamma_{c+1}(L) + L'')) = 0.$$

Functoriality of the dimension series now gives $q(\delta_{2c-1}(L)) = 0$, which proves the inclusion. For $c = 3$, the relation $L'' \subseteq \gamma_4(L)$ gives

$$\delta_5(L) \subseteq \gamma_4(L) + L'' = \gamma_4(L).$$

□

Corollary 5.8. Lean 4
If L is metabelian, then

$$\delta_{2c-1}(L) \subseteq \gamma_{c+1}(L), \qquad c \geq 2.$$

Proof. This follows from Corollary 5.7, since $L'' = 0$. □

Corollary 5.9. Lean 4
Let $c \geq 2$. If L is nilpotent of class at most c, then

$$\delta_{2c-1}(L) \subseteq L'' \cap Z(L).$$

Proof. Corollary 5.7 gives $\delta_{2c-1}(L) \subseteq \gamma_{c+1}(L) + L'' = L''$. Theorem B gives

$$[\delta_{2c-1}(L), L] = \gamma_{2c}(L) = 0,$$

so $\delta_{2c-1}(L)$ is central. □

Corollary 5.10. Lean 4
For every metabelian Lie ring L,

$$\bigcap_{n\geq 1} \delta_n(L) = \bigcap_{n\geq 1} \gamma_n(L).$$

Proof. The inclusion from right to left follows from (1). Conversely, Corollary 5.8 gives

$$\bigcap_{n\geq 1} \delta_n(L) \subseteq \delta_{2c-1}(L) \subseteq \gamma_{c+1}(L)$$

for every $c \geq 2$. Hence

$$\bigcap_{n\geq 1} \delta_n(L) \subseteq \bigcap_{c\geq 2} \gamma_{c+1}(L) = \bigcap_{n\geq 1} \gamma_n(L),$$

which proves the reverse inclusion. □

6. Formal verification

Theorems A, B, and C, together with the structural lemmas used in their proofs, have been verified in Lean 4. The source code is available in [**?**].

The formalization includes the integral PBW theorem in the form used in Section 2, including the injectivity of $L \to \mathrm{U}(L)$. It also includes the results

$$\delta_2(L) = \gamma_2(L), \qquad \delta_3(L) = \gamma_3(L), \qquad 2\delta_4(L) \subseteq \gamma_4(L),$$

and the equality of the two series for free Lie rings, following [**?**].

References